OS Orange level Problem 2

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Section – F

Phase 1 -

SS 1A –

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

SS 1B –

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

Phase 2-

SS 2A -

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ ./test_tree
Serialized tree: 140 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

SS 2B-

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ find .pes/objects -type f
.pes/objects/23/7b7cbf8cf88b36d349b368503cd9cd1fc28f3e058c414994a95e860e3c1495
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ xxd .pes/objects/23/7b7cbf8cf88b36d349b36
8503cd9cd1fc28f3e058c414994a95e860e3c1495
00000000: 626c 6f62 2033 3500 4669 6e61 6c20 7465  blob 35.Final te
00000010: 7374 2066 6f72 2070 6861 7365 2032 6220  st for phase 2b
00000020: 7363 7265 656e 7368 6f74 0a              screenshot.
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

Phase 3 –

SS3A-

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ ./pes add file1.txt
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ ./pes status
Staged changes:
  staged:   test.txt
  staged:   file1.txt

Unstaged changes:

Untracked files:
  untracked: commit.c
  untracked: commit.h
  untracked: commit.o
  untracked: index.c
  untracked: index.h
  untracked: index.o
  untracked: Makefile
  untracked: object.c
  untracked: object.o
  untracked: pes.c
  untracked: pes.h
  untracked: pes.o
  untracked: README.md
  untracked: test2b.txt
  untracked: test_objects
  untracked: test_objects.c
  untracked: test_objects.o
  untracked: test_sequence.sh
  untracked: test_tree
  untracked: test_tree.c
  untracked: test_tree.o
  untracked: tree.c
  untracked: tree.h
  untracked: tree.o
```

SS 3B-

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ cat .pes/index
100777 1b6b8dbde5c763eeb3bcced502fceaee1cc14fa0a9914f32bf28d9cbfce89372 1776688684 23 test.txt
100777 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776689493 6 file1.txt
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

Phase 4-

SS 4A -

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ ./pes log
commit 655c81d34783211a3ba7599f96e9842510d9ba283c42856b6a5b2db1b9bc1596
Author: ◆
Date:   1769472

    Third Commit


commit e236b2dd6c2e6f825b437e8cf54467906d8ea734457fcfe5107825b3f3b72d7e
Author: ◆
Date:   1769472

    Second Commit


commit 6798a7c4367b6b6dcedaad3116f210d6904f34ba28bcbb22d8c8a7de39e11121
Author: ◆
Date:   1769472

    First Commit


dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

SS 4B -

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ find .pes/objects -type f | sort
.pes/objects/27/10f8b6e4a35a418749361b448034de981d66a94b98f26b29ab8b4b83b0b2d4
.pes/objects/65/5c81d34783211a3ba7599f96e9842510d9ba283c42856b6a5b2db1b9bc1596
.pes/objects/67/98a7c4367b6b6dcedaad3116f210d6904f34ba28bcbb22d8c8a7de39e11121
.pes/objects/9f/257516457eddf8f5bd11fc32c4c9c25de63e4fb189454e3c94edad066d1b35
.pes/objects/ab/dae192663f6931681670d352aa6a735bee72e66ac0e75f57c6169a7046ebd5
.pes/objects/cf/93adb158f7aa63a85c53813a7292057de30727f767ef12298969439c9f0054
.pes/objects/e2/36b2dd6c2e6f825b437e8cf54467906d8ea734457fcfe5107825b3f3b72d7e
.pes/objects/e9/008cd294686d9c57a7bd368f71bf3731ff49563cdbe438459001e62aff929d
.pes/objects/fe/9216db872a471c5c014ffb150c181ae3d72a4aa26690daab574c2832ca14e9
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

SS 4C-

```
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$ cat .pes/refs/heads/main
655c81d34783211a3ba7599f96e9842510d9ba283c42856b6a5b2db1b9bc1596
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/OS ppt/assignment 4/PESXUG24CS824-pes-vcs$
```

Final –

```
dminshreyas@Shreyas46:~/pes_project$ make test-integration
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:   file.txt
  staged:   hello.txt

Unstaged changes:

Untracked files:

--- First Commit ---
[main ff416e175b212f4fdbc02d6b56c4b5b2b5a6f71453e7aed1dac0f795b506fd04] Initial commit
Committed: ff416e175b21... Initial commit

Log after first commit:
commit ff416e175b212f4fdbc02d6b56c4b5b2b5a6f71453e7aed1dac0f795b506fd04
Author: ◆
Date:   1769472

    Initial commit
```

```
!


--- Second Commit ---
[main 3e95d905df4be7654650da653309eb3efd018a9bdc96917ab60defdc4b410bb1] Update file.txt
Committed: 3e95d905df4b... Update file.txt

--- Third Commit ---
[main dc9a2b9edba8690d754976c9de67373110656e1b9170b0b1b9d26d98f21e5e70] Add farewell
Committed: dc9a2b9edba8... Add farewell

--- Full History ---
commit dc9a2b9edba8690d754976c9de67373110656e1b9170b0b1b9d26d98f21e5e70
Author: ◆
Date:   1769472

    Add farewell


commit 3e95d905df4be7654650da653309eb3efd018a9bdc96917ab60defdc4b410bb1
Author: ◆
Date:   1769472

    Update file.txt


commit ff416e175b212f4fdbc02d6b56c4b5b2b5a6f71453e7aed1dac0f795b506fd04
Author: ◆
Date:   1769472

    Initial commit
!
```

```
--- Reference Chain ---
HEAD:
dc9a2b9edba8690d754976c9de67373110656e1b9170b0b1b9d26d98f21e5e70
refs/heads/main:
dc9a2b9edba8690d754976c9de67373110656e1b9170b0b1b9d26d98f21e5e70

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/3e/95d905df4be7654650da653309eb3efd018a9bdc96917ab60defdc4b410bb1
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/79/09809c42e0815da81f57dc18c4505e17f7faec9054084c41b835326bb1a254
.pes/objects/89/eb6ed2608d3c0da045446ce8d41095deb627111906c7988d417ff67ea45ae3
.pes/objects/96/a21a5d3be2cda9c1dca016fe9c764ba577f26e44398e954a61497fb91c83b0
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/dc/9a2b9edba8690d754976c9de67373110656e1b9170b0b1b9d26d98f21e5e70
.pes/objects/ff/416e175b212f4fdbc02d6b56c4b5b2b5a6f71453e7aed1dac0f795b506fd04

=== All integration tests completed ===
dminshreyas@Shreyas46:~/pes_project$
```

Analysis questions -

Phase 5: Branching Analysis

Q5.1: How does creating a new branch in a VCS like Git (or PES-VCS) differ from creating a full copy of the project directory?

Answer: Creating a branch is a **metadata operation**, whereas copying the project is a **filesystem operation**. In PES-VCS, a branch is simply a small text file (a reference) containing a 64-character commit hash. It points to an existing commit in the object store. A full project copy duplicates every file and folder, consuming significant disk space ($O(n)$ where n is total data size). Branching consumes negligible space ($O(1)$) and allows for near-instantaneous switching between development lines.

Q5.2: In our implementation, we used `.pes/refs/heads/main` to track the current commit. If we wanted to create a branch named `feature`, what steps would the VCS take internally?

Answer: Internally, the VCS would perform the following steps:

1. Read the current commit hash from `.pes/HEAD` or `.pes/refs/heads/main`.
2. Create a new file at `.pes/refs/heads/feature`.
3. Write that same commit hash into the new `feature` file.
4. To "switch" to the branch, the `.pes/HEAD` file would be updated to point to `refs/heads/feature` instead of `main`.

Q5.3: What is a "detached HEAD" state, and how would our current `commit_create` logic need to change to support it?

Answer: A "detached HEAD" occurs when `.pes/HEAD` contains a **direct commit hash** rather than a reference to a branch name. In this state, you are looking at a specific point in history rather than the tip of a branch. To support this, our `commit_create` logic would need to detect if `HEAD` is a ref or a hash. If it's a hash, a new commit would update the `HEAD` file directly with the new hash, rather than updating a branch file in `/refs/heads/`.

Phase 6: Garbage Collection (GC) Analysis

Q6.1: Looking at your `find .pes/objects` output, if you were to delete a branch, why don't the objects (blobs/trees) disappear immediately?

Answer: This is due to the **content-addressed storage** model. Objects are stored independently of the branches that point to them. Deleting a branch only deletes the pointer (the "ref" file). The objects remain in the store because the VCS does not perform an automatic "check-on-delete" to see if other branches or commits still

reference those objects. They become "orphaned" or "dangling" but stay on disk until a manual Garbage Collection (GC) is triggered.

Q6.2: Describe an algorithm for a `pes gc` command that would safely remove unused objects without deleting active project data.

Answer: A safe GC algorithm would use a **Mark-and-Sweep** approach:

1. **Mark Phase:** Start a traversal from all "known roots" (all files in `.pes/refs/heads/` and the `.pes/HEAD` file). For every commit hash found, recursively mark the associated tree, its sub-trees, and all blobs.
2. **Identify:** Compile a list of every object hash currently present in `.pes/objects/`.
3. **Sweep Phase:** Compare the list of existing objects against the "Marked" list. Any object hash not found in the "Marked" list is considered unreachable (dangling) and can be safely deleted from the filesystem.

Github link -

https://github.com/Shreyas37-p/PESXUG25CS824-pes-vcs-.git